

Backend Overview

The backend delivers safe and scalable solutions for:

- HTTP Basic and Digest Access
- User profile management (language background, level, objectives, learning styles)
- Exercise delivery + adaptive difficulty logic (retrieve questions, evaluate exercise attempts, update difficulty)
- AI assistant services (translate, explain, practice generation)
- Progress tracking and analytics (learner data and teacher data)
- Notifications (reminders, new exercises, messages from teachers)

Backend Architecture

1. API Gateway / Backend Application
 - Node.js (TypeScript) or Python (FastAPI).
 - Exposes REST endpoints at: /auth, /profile, /exercise, /assistant, /analytics.
2. Primary Database (PostgreSQL)
 - Stores users, profiles, exercises, attempts, progress, vocabulary, relations between teachers and classes
3. Event Logs Store (Firebase, MongoDB)
 - Stores high-volume events such as screen views, clicks, assistant usage, latency measurements, etc.
4. AI/ML Layer (as a separate service or component)
 - Translation / explanation / practice generation
 - Difficulty prediction / recommendation scoring
 - Prediction of engagement prediction for teacher analytics.